Answers

A.1

Gold, silver, and carbon are all examples of elements. They are listed on the periodic table of elements. If you had a golden ring and an amazing machine, you could break it down into tiny pieces called atoms.

Isotope	**Atomic Symbol**	**Number of Protons**	**Number of Neutrons**
Uranium-238	U	92	146
Radium-226	Ra	88	138
Lead-206	Pb	82	124
Hydrogen-3	H	1	2
Sodium-22	Na	11	11
Argon-39	Ar	18	21
Radon-222	Rn	86	136
Carbon-12	C	6	6
Carbon-13	C	6	7
Carbon-14	C	6	8

Carbon-12, Carbon-13, and Carbon-14 are all different isotopes of carbon.

B.1. $^{23}_{11}Na$

B.2. $^{16}_{8}O$

B.3. $^{1}_{1}H$

B.4. $^{124}_{52}Te$

B.5. $^{40}_{18}Ar$

Nucleus	Element	Number of Protons	Number of Neutrons

$^{14}_{7}N$	Nitrogen	7	7
$^{1}_{1}H$	Hydrogen	1	0
$^{222}_{86}Rn$	Radon	86	136
$^{238}_{92}U$	Uranium	92	146
$^{14}_{6}C$	Carbon	6	8

C.1 $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$

C.2 $^{222}_{88}Ra \rightarrow {}^{218}_{86}Rn + {}^{4}_{2}He$

C.3 $^{208}_{84}Po \rightarrow {}^{204}_{82}Pb + {}^{4}_{2}He$

D.1 gamma ray

D.2 alpha particle

D.3 beta particle

D.4 beta particle

D.5 alpha particle

D.6 alpha particle

D.7 beta particle

Draw an alpha decay for each nuclide.

	Nuclide		Decay Reaction
D.8	$^{256}_{103}Lr$		$^{256}_{103}Lr \rightarrow {}^{252}_{101}Md + {}^{4}_{2}He$
D.9	$^{231}_{91}Pa$		$^{231}_{91}Pa \rightarrow {}^{227}_{89}Ac + {}^{4}_{2}He$
D.10	$^{225}_{89}Ac$		$^{225}_{89}Ac \rightarrow {}^{221}_{87}Fr + {}^{4}_{2}He$